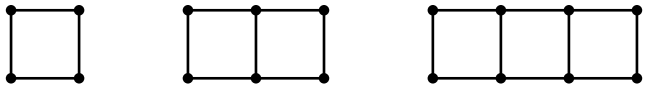
Linear Number Patterns – Extra Practice

These questions are extra practice in the same style as Questions 1 and 2 of the *Linear number patterns* worksheet: a picture pattern grows by the same amount each step, so the number of items follows a linear rule. For each question: (a) draw the next pattern, (b) complete the table, (c) find a formula, then (d)–(e) use your formula to answer questions about much larger patterns.

Question 1 – Matchstick Squares

This pattern shows the number of matchsticks needed to make a row of joined squares.



(a) Draw the next pattern (4 squares in a row).

Draw your pattern here.

(b) Copy and complete the following table.

Number of squares (S)	1	2	3	4	5	12
Number of matches (M)						

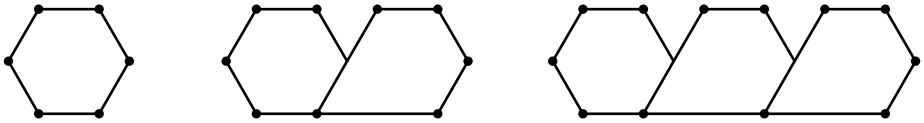
(c) Construct a formula to find the number of matches needed for any number of squares.

(d) If a row had 40 squares, how many matches would be needed?

(e) If a row had 75 squares, how many matches would be needed?

Question 2 – Hexagonal Picnic Tables

A campsite joins hexagonal picnic tables end to end in a row, as shown below (each dot is a seat). When two tables are pushed together, the shared side can no longer seat anyone.



(a) Draw the next pattern (4 hexagonal tables joined in a row).

Draw your pattern here.

(b) Copy and complete the table below.

Number of tables (T)	1	2	3	4	5	25
Number of seats (C)						

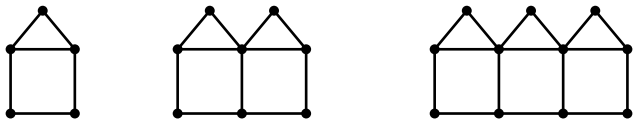
(c) Construct a formula for the number of seats needed for any number of tables.

(d) If the campsite joined 40 tables in a row, how many seats would there be?

(e) If the campsite needed to seat 70 people in a single row of joined tables, how many tables would be needed?

Question 3 – Matchstick Houses

This pattern shows the number of matchsticks needed to make a row of joined "houses" (a square with a triangular roof). Neighbouring houses share one wall.



(a) Draw the next pattern (4 houses in a row).

Draw your pattern here.

(b) Copy and complete the following table.

Number of houses (H)	1	2	3	4	5	15
Number of matches (M)						

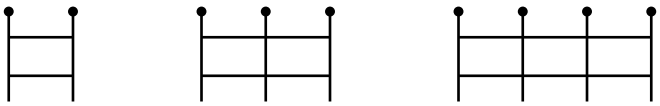
(c) Construct a formula to find the number of matches needed for any number of houses.

(d) If a row had 50 houses, how many matches would be needed?

(e) If a row had 80 houses, how many matches would be needed?

Question 4 – Garden Fencing

A fence is built from posts with two horizontal rails running between each pair of neighbouring posts, as shown below.



(a) Draw the next pattern (5 posts).

Draw your pattern here.

(b) Copy and complete the table below.

Number of posts (P)	2	3	4	5	6	7
Number of rails (R)						

(c) Construct a formula for the number of rails needed for any number of posts.

(d) If a fence had 50 posts, how many rails would be needed?

(e) If a fence used 200 rails, how many posts would it have?